

Experiment : 7

Caching Application Data with ElastiCache, Caching with Amazon CloudFront, Caching Strategies

1. Caching Application Data with ElastiCache

Amazon ElastiCache

- It stores **data in memory (RAM)**
- Used by applications to get data **very fast**

What is it?

Saving frequently used data (like user info, results) in cache instead of database.

2. Caching with CloudFront

Amazon CloudFront

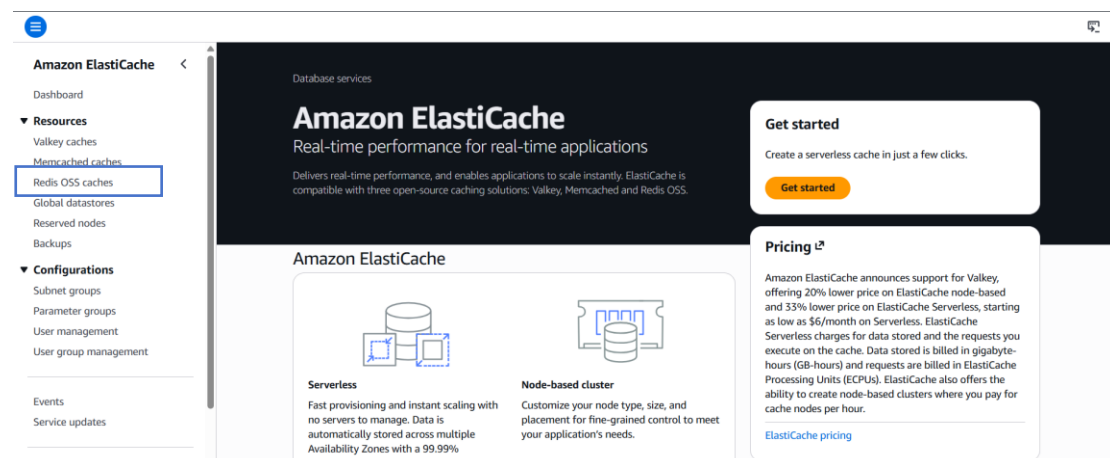
- It is a **Content Delivery Network (CDN)**
- Stores content closer to users (edge locations)

What is it?

Caching website content (images, HTML, videos) near users for fast delivery.

Step:1

- Go to AWS Management Console
- Login with your AWS account
- In the top search bar, type **ElastiCache**
- Click on Amazon ElastiCache from the services list



OSS. Step 2: Configure Redis Cluster

DO NOT enable Multi-AZ.

Select Engine Version: 7.0.

Set Port Number: 6379.

Choose Node Type: t3.micro.

Set Shards: 2.

Provide a Redis Subnet Name and manage the resource.

The image displays three sequential screenshots of the AWS ElastiCache console during the 'Create cache' process. The first screenshot shows the 'Settings' step, where 'Redis OSS' is selected as the engine and 'Node-based cluster' as the deployment option. The second screenshot shows the 'Creation method' step, where 'Cluster cache' is selected and 'Cluster mode' is set to 'Enabled'. The third screenshot shows the 'Cluster info' step, where the cluster name is 'newcache' and the location is 'AWS Cloud'. The 'Multi-AZ' option is also visible and selected.

Step 1 Settings

Configuration Info

Choose one of the following options to create a new cache.

You can save up to 33% with Serverless and 20% with node-based ElastiCache by choosing Valkey. Valkey is open source, established under the Linux Foundation, and fully compatible with Redis OSS v7.0.

Engine

☐ Valkey - recommended ☐ Memcached ☒ Redis OSS

Valkey offers enhanced price and performance while staying true to its open source roots.
Valkey is an open source, in-memory, high performance, key-value datastore and a drop-in replacement for Redis OSS. Start using Valkey today and learn about its benefits. [Learn more](#)

Deployment option

☐ Serverless
Use to quickly create a cache that automatically scales to meet application traffic demands, with no servers to manage.

☒ Node-based cluster
Use to create a cache by selecting node type, size, and count.

Creation method

☐ Easy create
Use recommended best practice configurations. You can also modify options after you create the cluster.

☒ Cluster cache
Set all of the configuration options for your new cluster.

☐ Restore from backup
Use an existing backup or .rdb file to restore a cluster.

Creation method

☐ Easy create
Use recommended best practice configurations. You can also modify options after you create the cluster.

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Cluster mode

Scale your cluster dynamically with no downtime.

☒ Enabled
Cluster mode enables replication across multiple shards for enhanced scalability and availability.

☐ Disabled
The Redis OSS cluster will have a single shard (node group) with one primary node and up to 5 read replicas.

Enabling cluster mode supports partitioning your data across up to 500 node groups and improves performance of Redis OSS clusters. Some commands are unavailable in this mode. [Learn more](#)

Cluster info

Use the following options to configure the cluster.

Name

newcache

The name can have up to 40 characters, and must not contain spaces.

Cluster info

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Name

newcache

The name can have up to 40 characters, and must not contain spaces.

Description - optional

Description

Location

Choose whether to host the cluster in the AWS Cloud or on premises.

Location

☒ AWS Cloud
Use the AWS Cloud for your ElastiCache instances.

☐ On premises
Create your ElastiCache instances on an Outpost (through AWS Outposts). You need to create a subnet ID on an Outpost first.

Multi-AZ

☒ Enable
Multi-AZ provides enhanced high availability through automatic failover to a read replica, cross AZs, in case of a primary node failure.

Auto-failover

7.1

Port
The port number that nodes accept connections on.
6379

Parameter groups
Parameter groups control the runtime properties of your nodes and clusters.
default.redis7.cluster.on

Node type
The type of node to be deployed and its associated memory size.
cache.t3.medium
3.09 GiB memory Up to 5 Gigabit network performance

Number of shards
Enter the number of shards in this cluster, from 1 to 500.
2

Replicas per shard
Enter the number of replicas for each shard, from 0 to 5.
1

Connectivity

CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Connectivity
Choose the IP version(s) this cluster will support. Then select an existing subnet group or create a new one.

Network type
Choose between IPv4, dual stack and IPv6.
IPv4
Your resources will communicate only over the IPv4 protocol.

Subnet groups
☐ Choose existing subnet group ☒ Create a new subnet group

Name
availablezone
The name can have up to 255 characters, and must not contain spaces.

Description - optional
Description

VPC ID
The identifier for the VPC environment where your cluster is to run.
vpc-05c83c79deab0f7c [Create VPC](#)

☒ For Multi-AZ high availability mode, choose IDs for at least two subnets from two Availability Zones in the table below.

Availability Zone placements
Use the following fields to configure placements for Availability Zones.

Slots and keyspaces
Distribution of the 16,384 cluster keypace slots across shards.
Equal distribution

Availability Zone placements
By locating nodes in different Availability Zones, you reduce the chance that a failure in one Availability Zone, such as a power outage, will cause your entire system to fail. Choose **Specify Availability Zones** if you want to specify Availability Zones for cluster nodes.
Specify Availability Zones

< 1 >

Shards	Slots/keyspaces	Primary	Replica 1
Shard 1	Equal distribution	us-east-1a	us-east-1b
Shard 2	Equal distribution	us-east-1a	us-east-1b

Cancel Next

Step 3: Configure Authentication

Select AUTH DEFAULT ACCESS.
Set Auth Token: AUTHtoken123456789.

Auth token
The AUTH token used for the clusters.

CloudCse@skit ☒ Show token

Password strength Very strong
At least 16 characters and a maximum of 128 characters, which can be any printable ASCII character except for " (blank space), ' (quotation mark), /, and @.

Selected security groups (0)
A security group acts like a firewall that controls network access to your clusters.

No selected security groups
Add security groups by choosing the Manage button.

Backup
You can use backups to restore a cluster or seed a new cluster. The backup consists of the cluster's metadata, along with all of the data in the cluster.

☒ Enable automatic backups
ElastiCache will automatically create a daily backup of a set of replicas.

Backup retention period

Step 4: Create a Security Group in EC2

Open a new tab and go to EC2 Dashboard.
Click Security Groups > Create Security Group.

Add Inbound Rule:

Type: Custom TCP

Port: 6379

Source: Anywhere (0.0.0.0/0 for IPv4)

Amazon Elastic Compute Cloud (EC2)
Create, manage, and monitor virtual servers in the cloud.

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.

Launch a virtual server

Additional actions

[View running instances](#)
[Migrate a server](#)

Pricing (US)

[EC2 pricing options](#)
[Use the AWS pricing calculator](#)

Benefits and features

EC2 offers ultimate scalability and control
Fully resizable compute capacity to support virtually any workload. This service is best if you want:

- Highest level of control of the entire technology stack, allowing full integration with all AWS services
- Widest variety of server size options
- Widest availability of operating systems to choose from including Linux, Windows, and macOS
- Global scalability

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
Custom TCP	TCP	6379	Anywh... 0.0.0.0/0		Delete

[Add rule](#)

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional	
All traffic	All	All	Custom 0.0.0.0/0		Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Step 5: Attach Security Group to ElastiCache

Go back to ElastiCache.

Select your Redis Cluster.

Click Manage and Refresh.

Select the Security Group created in EC2.

Manage security groups

Security groups (2)

☐	Security group ID	Security group name	Description
☒	sg-0be48f3c5103c69d1	new_security_group	this new group
☐	sg-0dd930214270f9399	default	default VPC security group

[Cancel](#) [Choose](#)

Elasticache > Caches > Create cache

Step 1: Settings

Cluster info
Use the following options to configure the cluster.

Name	newcaches	Description	-
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Location
Choose whether to host the cluster in the AWS Cloud or on premises.

Location	AWS Cloud	Cluster mode	Enabled
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Cache settings
Use the following options to configure the cluster.

Engine	Redis	Engine version	7.1	Port	6379
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Step 6: Finalize and Create Redis Cache

Keep the default settings on the next page.

Click Create to deploy Redis Cache.

us-east-1.console.aws.amazon.com/elasticache/home?region=us-east-1#/redis

The cluster was created successfully.

Amazon Elasticache announces Valkey starting as low as \$6/month
Introducing Valkey, an open source project that is a drop in replacement of Redis OSS, now available on Elasticache, with up to 33% lower price than other engines. [Learn more](#)

Redis OSS caches (1)

Cache name	Status	Description	Engine version	Configuration	Created date
newcaches	Available	-	7.1.0	cache.t3.medium	April 5, 2026, 22:01:24...